

NICKOLAS J. VELA

US Citizen | 408-603-4017 | niccjvela27@gmail.com | www.linkedin.com/in/nickvela | nickvela.dev

EDUCATION

UNIVERSITY OF CALIFORNIA, SANTA CRUZ

Santa Cruz, CA

B.S. Computer Science

Expected Jun 2027

Cumulative GPA: 3.94/4.00

Relevant Coursework: Data Structures & Algorithms, Artificial Intelligence, Machine Learning, Computer Systems and C Programming, Python Programming, Computer Architecture, Geospatial Systems, Linear Algebra, Multivariable Calculus

EXPERIENCE

OVERTURE MAPS FOUNDATION

Hybrid

Lead Research Intern

Jan 2026 – March 2026

- Built an AI pipeline combining YOLOv8 street-view detection with a ResNet-18 satellite CNN fallback to predict building entrance locations, increasing entrance coverage from 5.8% to 56% across a 500-point national ground-truth dataset.
- Trained a satellite fallback model on 500 verified entrance points that activates automatically when street-level imagery is unavailable, achieving lower median prediction error than the street-view-only baseline while covering 10x more locations.
- Presented research findings to engineers and leadership from Overture Maps Foundation sponsor companies including Meta, Amazon, Microsoft, Niantic, and TomTom.
- Led a 4-person research team, directing project strategy, coordinating individual research approaches, and organizing the team's final presentation to sponsor company's engineers and leadership.
- Built a head-to-head accuracy evaluation framework comparing predictions against 500 manually verified ground-truth entrance points, with automated Haversine distance statistics and interactive Leaflet comparison maps.

UNIVERSITY OF CALIFORNIA, SANTA CRUZ

Santa Cruz, CA

Resident Assistant

Aug 2025 – Present

- Served as a community leader and contact point for 50+ residents, creating an inclusive and supportive living environment.
- Developed strong leadership and communication skills through weekly meetings, one-on-one mentoring, and facilitating group discussions.

PROJECTS

PROFESSIONAL PORTFOLIO WEBSITE — nickvela.dev

Oct 2025

Technologies: JavaScript, CSS, HTML

- Engineered a fully responsive and mobile-friendly website using HTML, CSS, and JavaScript, ensuring seamless performance across devices and screen sizes.
- Optimized user engagement through interactive elements including draggable physics objects, 3D CSS transforms with perspective effects, and debounced scroll event handlers for improved performance

RALLY — IOS EVENT & COMMUNITY APP

Oct 2025

Technologies: Swift, SwiftUI, Xcode, Firebase, Firestore

- Designed, developed, and deployed a fully functional iOS application that enables college students to create, join, and manage community events in real time.
- Implemented Firebase Authentication, Firestore real-time database, and Cloud Storage infrastructure with optimized queries achieving sub-100ms response times and enabling scalability to 10,000+ concurrent users.
- Developed admin dashboard with content moderation tools, multi-level reporting system, user management controls, and role-based access reducing moderation workflow time by 70%.
- Integrated MapKit and CoreLocation APIs to create location-based event discovery with coordinate search, interactive maps, and address autocomplete functionality for campus-wide event exploration.

ADDITIONAL

Languages: Python, C, C++, Swift, JavaScript, SQL, HTML, CSS, Shell Script

Technologies: SwiftUI, Bash, Git, Vim, Firebase, Firestore, VS Code, Xcode, Claude Code, DuckDB

Libraries/Frameworks: Pandas, NumPy, Scikit-Learn, Matplotlib, PyTorch, Ultralytics